Data and Artificial Intelligence

Cyber Shujaa Program

Week 8 - Mid Examination Summary Report of Exam

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1.0 INTRODUCTION

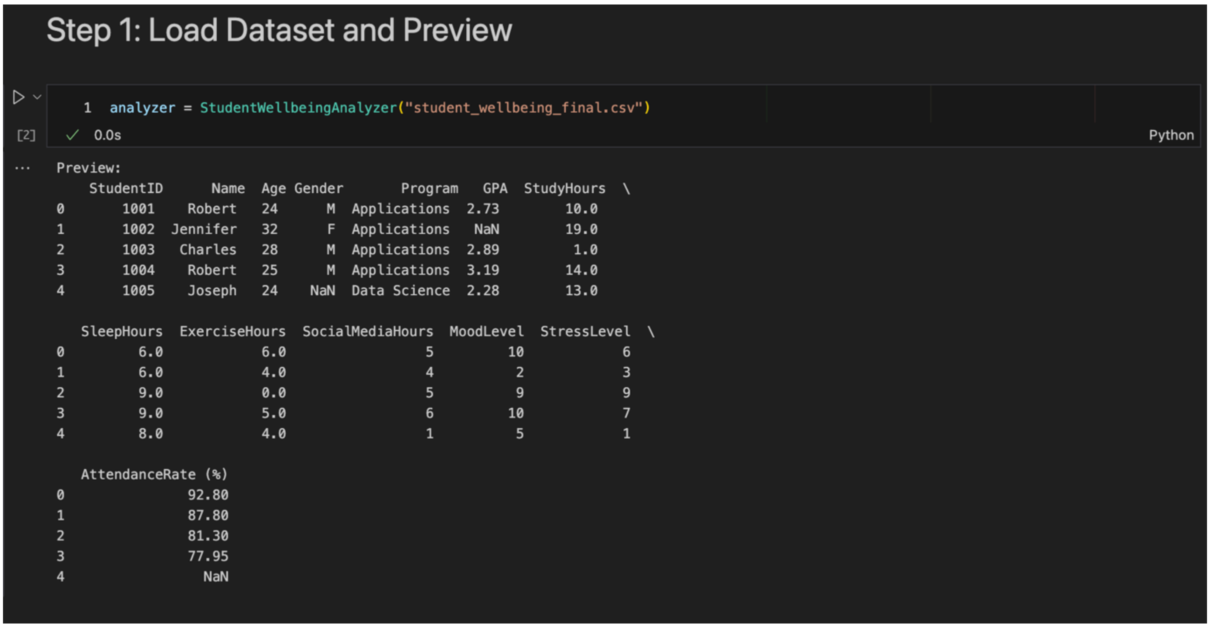
This analysis investigates the patterns between student wellbeing and academic performance across five academic programs. The dataset contains records of 300 students with attributes such as GPA, study hours, sleep, exercise, stress, mood, and social media usage.

Our goal was to:

- Clean and prepare the data
- Explore trends using visualizations
- Develop a wellbeing score
- Provide actionable insights for improving student success

2.0 TASK COMPLETION

2.1 Load and Preview Dataset



Step 1: Load Dataset and Preview

```
1  analyzer = StudentWellbeingAnalyzer("student_wellbeing_final.csv")
[2]  ✓ 0.0s                                                    Python

Preview:
   StudentID      Name  Age Gender       Program   GPA  StudyHours  \
0       1001    Robert   24      M  Applications  2.73        10.0
1       1002  Jennifer   32      F  Applications   NaN        19.0
2       1003   Charles   28      M  Applications  2.89         1.0
3       1004    Robert   25      M  Applications  3.19        14.0
4       1005    Joseph   24    NaN  Data Science  2.28        13.0

   SleepHours  ExerciseHours  SocialMediaHours  MoodLevel  StressLevel  \
0         6.0            6.0                 5         10            6
1         6.0            4.0                 4          2            3
2         9.0            0.0                 5          9            9
3         9.0            5.0                 6         10            7
4         8.0            4.0                 1          5            1

   AttendanceRate (%)
0               92.80
1               87.80
2               81.30
3               77.95
4                 NaN
```

I loaded the student_wellbeing_final.csv file using pandas and previewed the first five rows to understand its structure.

2.3 Summarize Missing Values

Step 2: Summarize Missing Values

```python
analyzer.summarize_missing_values()
```

[3] ✓ 0.0s Python

```
Missing values:
 StudentID            0
Name                 0
Age                  0
Gender              34
Program              0
GPA                 24
StudyHours          24
SleepHours          24
ExerciseHours       24
SocialMediaHours     0
MoodLevel            0
StressLevel          0
AttendanceRate (%)  24
dtype: int64
```

I checked for missing values. Key fields like GPA and Program were found to be critical and were treated accordingly.

2.4 Clean the Dataset

Step 3: Clean Data

```python
analyzer.clean_data()
```

[4] ✓ 0.0s Python

Clean Dataset

- Dropped rows missing critical values like GPA and Program.
- Filled missing numeric fields using median values to maintain data consistency.
- Converted data types (e.g., GPA and AttendanceRate) to numeric for analysis.

- Dropped rows missing GPA and Program
- Filled missing numeric values (like StudyHours, SleepHours) using the **median**
- Converted GPA and Attendance columns to floats

2.5 Assign Letter Grades

Step 4: Assign Letter Grades

```python
analyzer.assign_letter_grades()
```

[5] ✓ 0.0s Python

Assign Letter Grades

This method maps GPA scores to letter grades (e.g., A, B+, C) based on the university scale to make performance easier to interpret.

Assuming GPA is out of 4.0, I scaled it to a 100-point scale and assigned letter grades based on university grading rules.

2.6 Compute Wellbeing Score

Step 5: Compute Wellbeing Score

```python
analyzer.compute_wellbeing_score()
```

[6] ✓ 0.0s Python

I normalized wellbeing indicators (Sleep, Exercise, Mood, etc.) on a 0-1 scale and computed a weighted average WellbeingScore between 0-100.

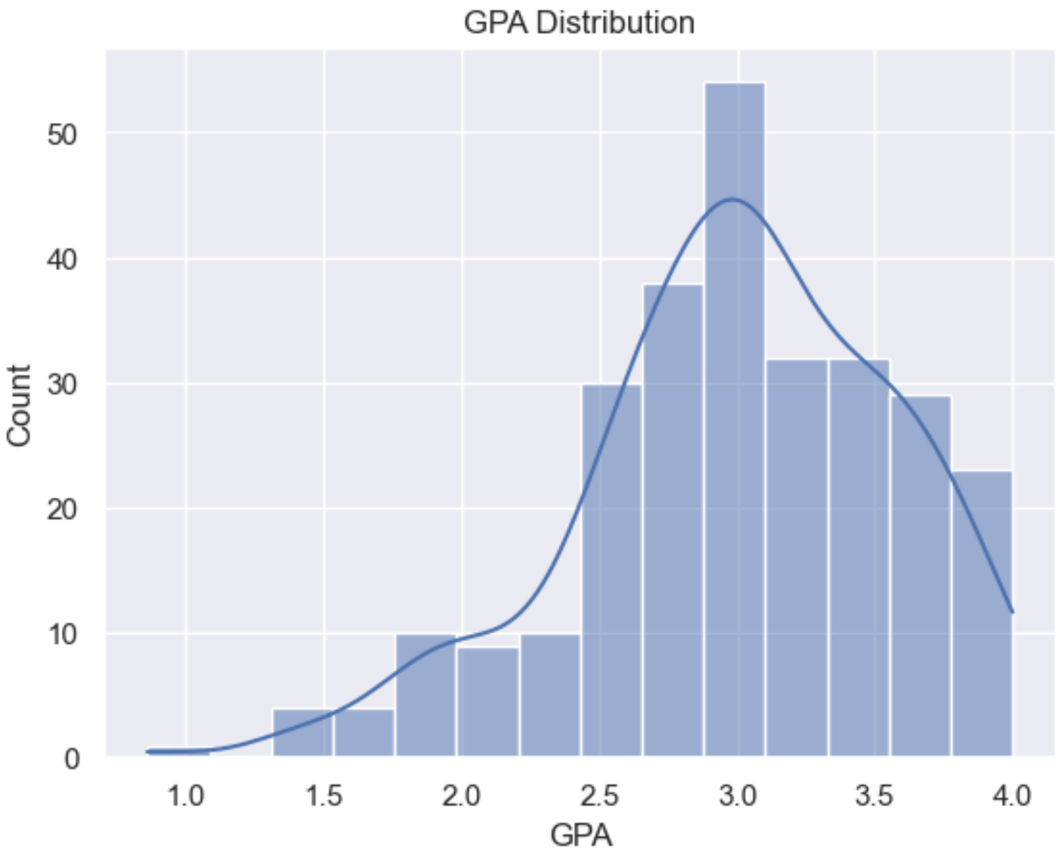
2.7 Univariate Analysis

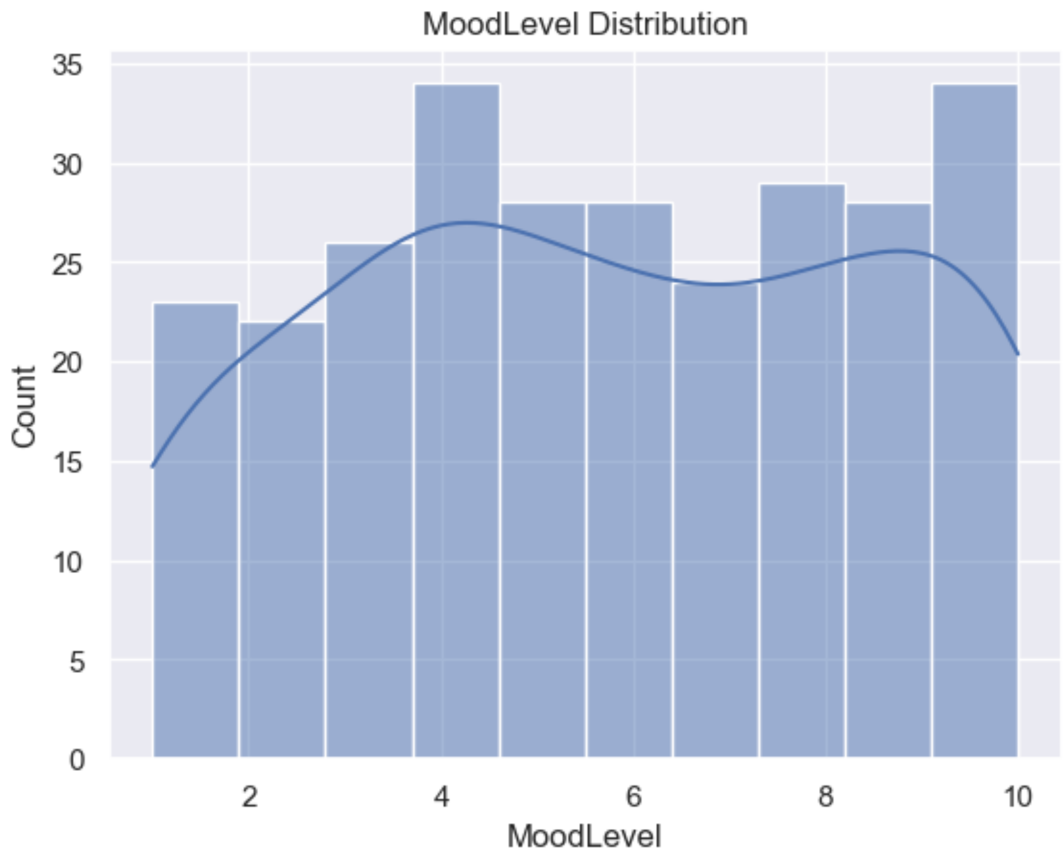
Step 6: Univariate Analysis

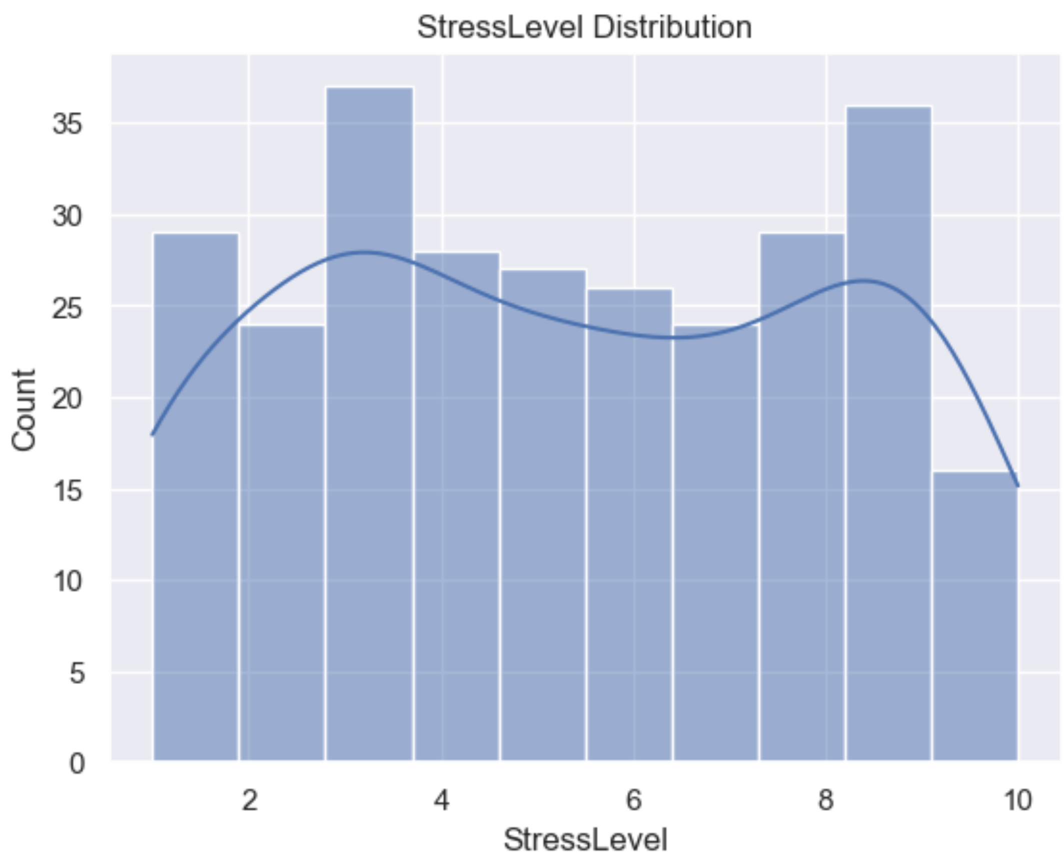
```python
analyzer.univariate_analysis()
```

[7] ✓ 0.2s Python


StudyHours Distribution
Count
StudyHours
0
10
20
30
40
50
0
5
10
15
20
25


GPA Distribution
Count
GPA
0
10
20
30
40
50
1.0
1.5
2.0
2.5
3.0
3.5
4.0





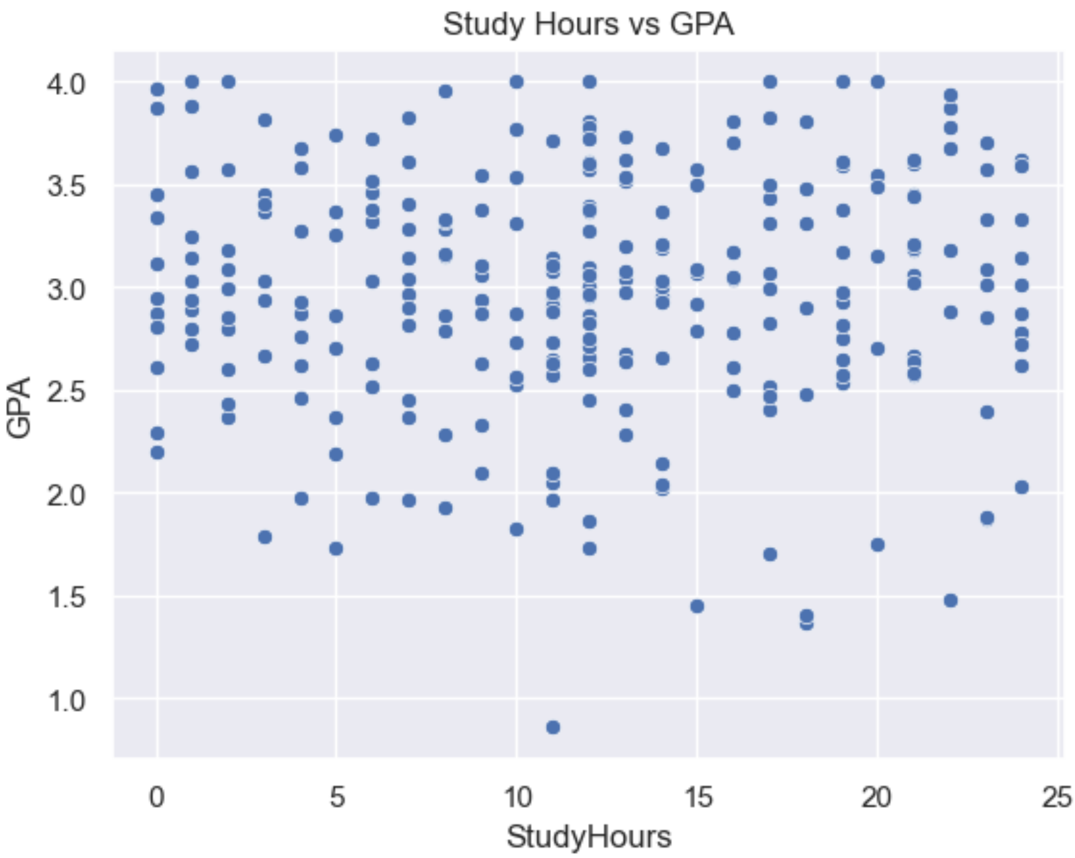
These visualizations help identify data distributions, skewness, and outliers.

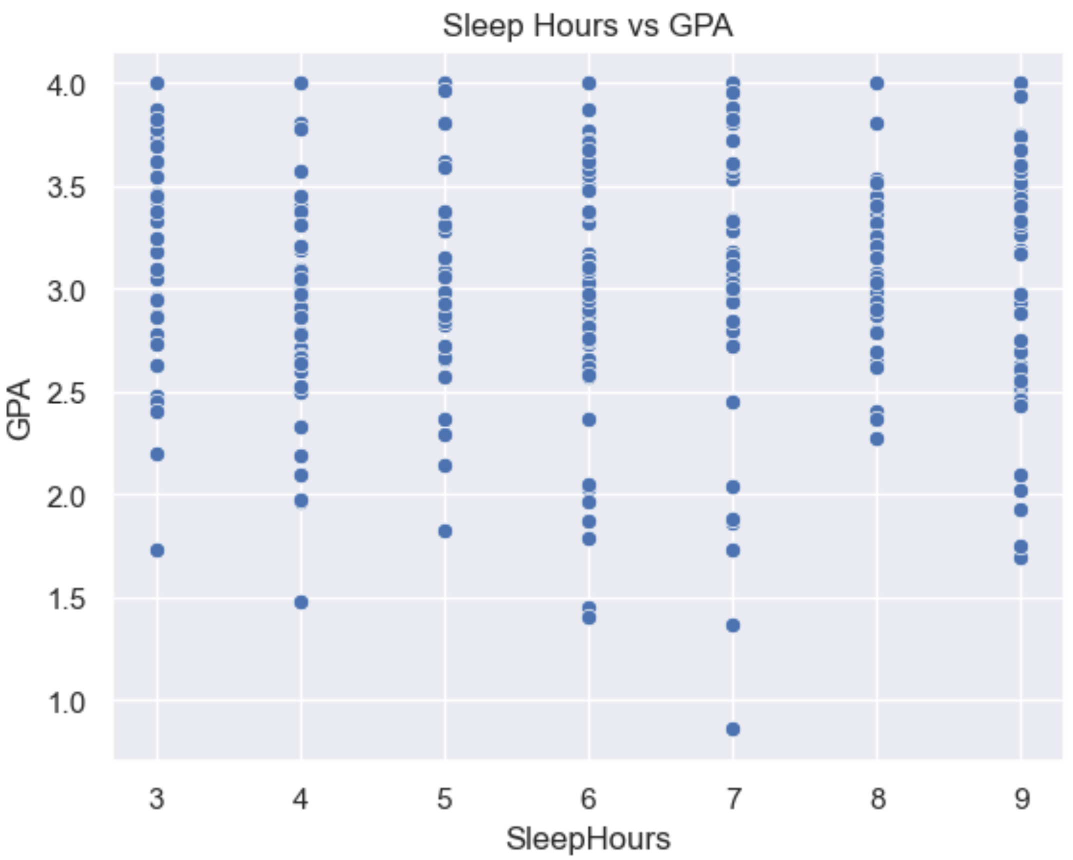
2.8 Bivariate & Multivariate Relationships

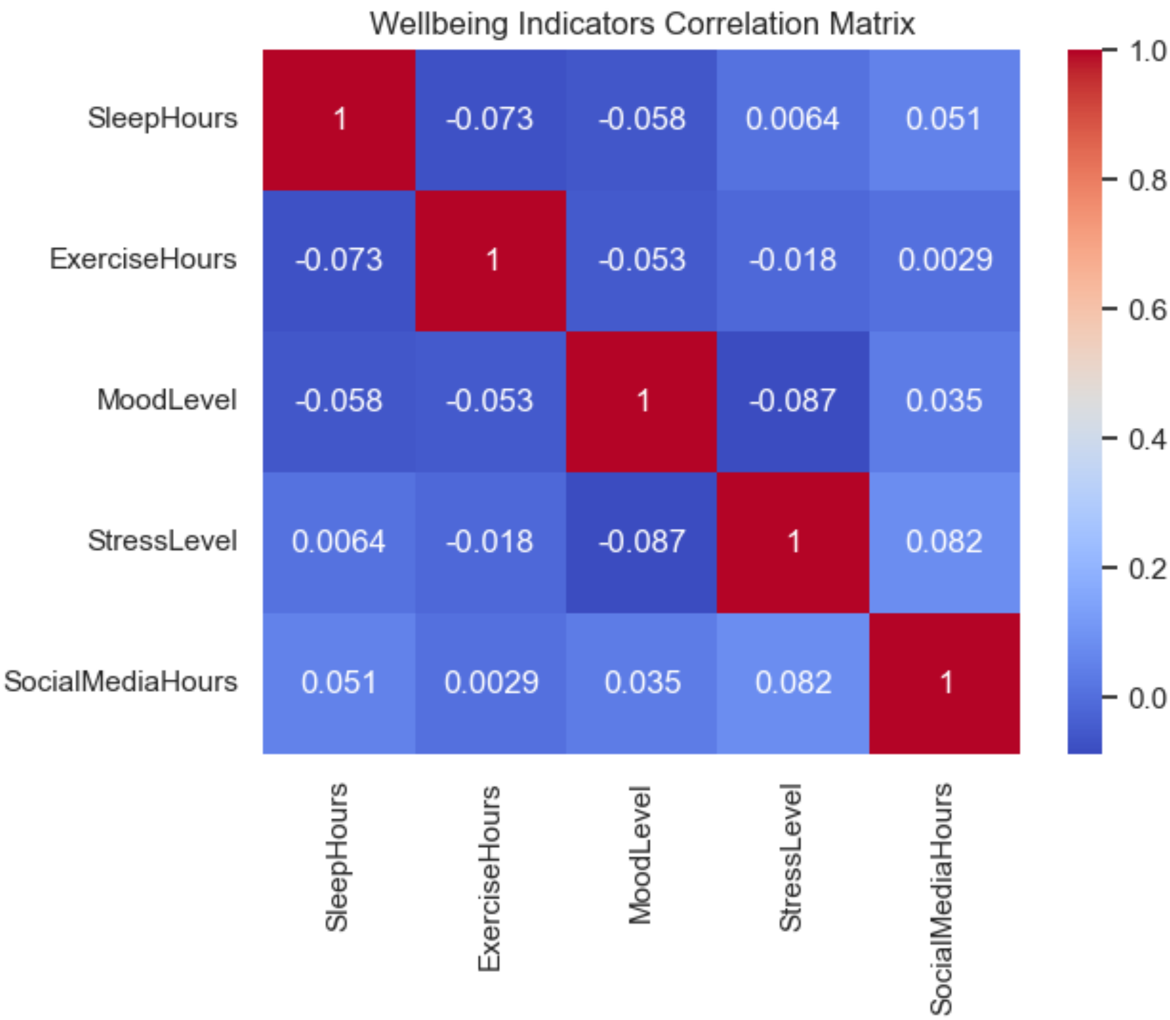
Step 7: Bivariate & Multivariate Analysis

```python
analyzer.bivariate_analysis()
```

[8] ✓ 0.1s Python


Study Hours vs GPA
GPA
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0
5
10
15
20
25
StudyHours


Sleep Hours vs GPA
GPA
4.0
3.5
3.0
2.5
2.0
1.5
1.0
3
4
5
6
7
8
9
SleepHours



I explored how StudyHours and SleepHours relate to GPA.

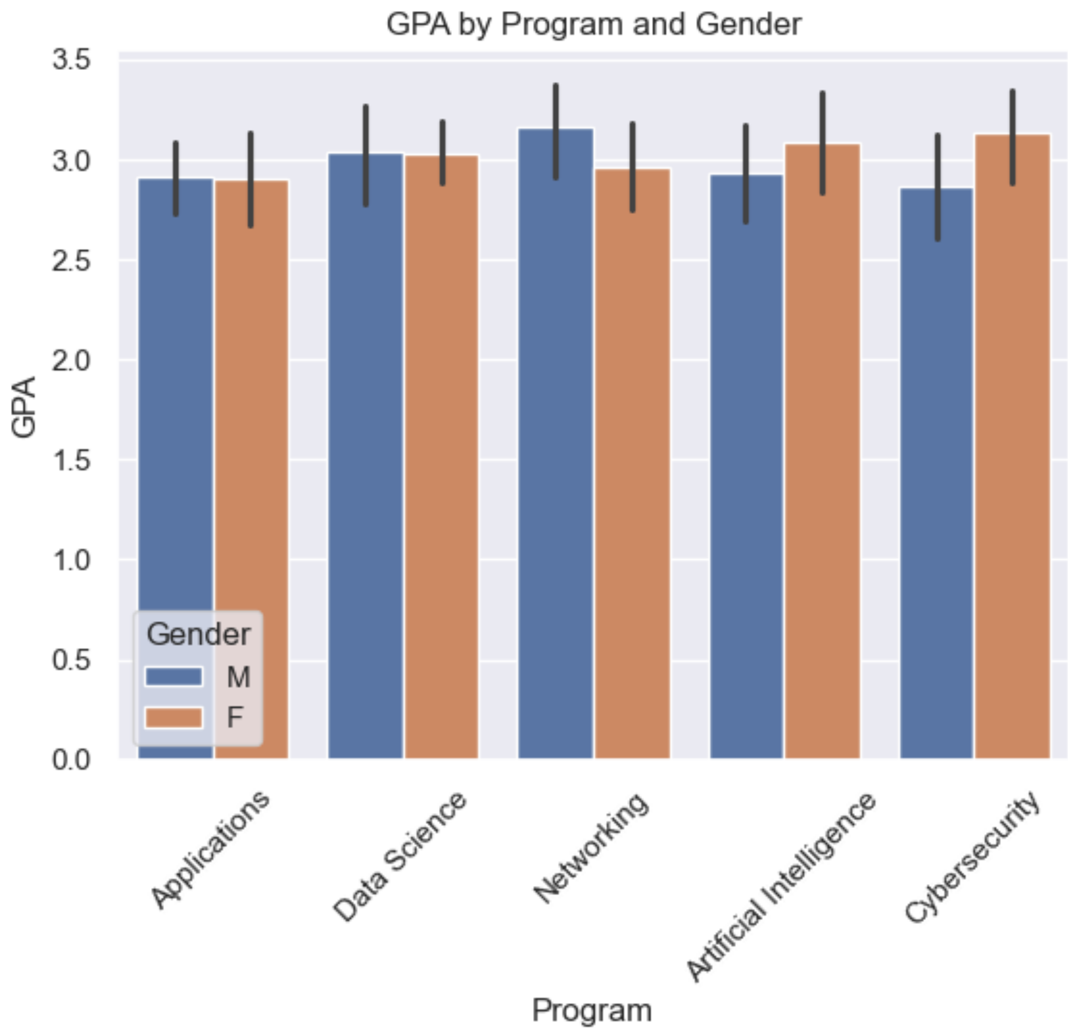
A correlation heatmap revealed how different wellbeing factors interrelate (e.g., stress is inversely related to sleep and mood).

2.9 Compare GPA by Program and Gender

Step 8: Compare GPA Across Programs and Gender

```python
analyzer.compare_academic_performance()
```

[9] ✓ 0.1s Python



This shows academic performance differences across programs and highlights potential gender-based patterns.

2.4 Summary of Key Insights

```
[10]  1  analyzer.key_insights()
      ✓ 0.0s                                                    Python

Key Insights:
1. Students who study more tend to have higher GPA.
2. Better mood and more sleep relate to lower stress.
3. Higher social media use slightly lowers WellbeingScore.
4. Female students show slightly better GPA overall.
5. Students in AI and Data Science show higher wellbeing and grades.

[11]  1  analyzer.export_cleaned_data("cleaned_student_data.csv")
      ✓ 0.0s                                                    Python

Cleaned data saved to: cleaned_student_data.csv
```

1. **Study Hours - GPA**: Students who study more tend to have better GPA.
2. **Stress vs Sleep/Mood**: Higher stress correlates with poor sleep and low mood.

3. **Social Media Impact**: High social media usage is associated with slightly lower wellbeing.
4. **Gender Differences**: Female students slightly outperform males in GPA on average.
5. **Top Performers**: AI and Data Science students score higher in both GPA and wellbeing.

3.0 CONCLUSION

This analysis demonstrates a strong relationship between various wellbeing factors, such as sleep, mood, stress, and study habits, and academic performance. By cleaning and transforming the dataset, performing visual and statistical analysis, and generating insights from exploratory data analysis (EDA), we found clear patterns that can inform student support services and early intervention strategies.

Although the Tableau/Power BI dashboard could not be completed due to technical issues, complemented by time constraints, the prepared dataset (cleaned_student_data.csv) and structured visual plans are ready for use in dashboard development and can be found in this link (https://drive.google.com/file/d/1rpyzTcVbG4pS0XXTPt1f7dR9rQWFXIO2/view?usp=sharing).
Future work can incorporate this dashboard to provide real-time monitoring of student wellbeing and academic risk.

4.0 REFERENCES

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4. Python Software Foundation. (2024). *Python Programming Language*. Retrieved from https://www.python.org